

Capstone Design Proposal
Optimization Business Flow Company Using ERP

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STATEMENT OF ORIGINALITY

In my capacity as an active student at President University and as the author of the Capstone Design Project stated below:

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I hereby declare that my Capstone Design Project entitled Optimization Business Flow using Enterprise Resource Planning will be personally responsible for the consequences of these acts of plagiarism and will accept the sanctions against these acts in accordance with the rules and policies of President University.

I also declare that this work, either in whole or in part, has not been submitted to another university to obtain a degree.

Cikarang, 2023

Signer 1	Signer 2	Signer 3
Student Name – NIM	Student Name – NIM	Student Name – NIM

TITLE

Approved:



Lecturers Name Capstone Advisor	{Head of Study Program Name} Program Head of XX

Prof. Dr. Ir. Wiranto Herry Utomo, M.Kom.
Dean of Faculty of Computer Science

BAB 1

A. PROBLEM SIMULATION

1. Background of the Problem

In the Business Environment right now, companies right now are required to have efficient, integrated, and well structured business processes to support operational activities and decision-making. However, the company is currently facing several challenges that relate to the business core flow, which is still not well organized and not fully integrated across department

At present, some operations activities are still carried out using separate systems or manual processes. As a result, data is stored in different platforms, leading to data redundancy delays in information exchange, and a higher risk of human error. These issues negatively impact coordination between departments such as warehouse and shipping, causing the overall business process to become less effective.

Furthermore, the absence of an integrated system limits management's ability to monitor business performance in real time, Decision-making processes often rely on incomplete or outdated information, which may affect the accuracy of strategic and operational decisions. This situation reduces the company's ability to respond quickly to operational problems operations

2. Problem from a Customer Perspective

From the customer perspective, currently business process flow often causes several issues. Such as insufficient data on incoming and outgoing goods, and tracking records it's cause the lack of system integration and unstructured workflows result in delays, inaccurate information, and limited transparency throughout the customer journey.

Specifically, customers may experience the following problems:

- **Delays in order processing and delivery**, due to inefficient coordinations between admin, warehouse, and logistics departments

- **Inaccurate product availability information**, where customers place orders for items that are actually out of stock.
- **Human errors occurs in order to fulfillment data**, such as incorrect items, quantities, or incomplete deliveries
- **Waste time to search data and collect data**, such as tracking delivery, receipt, and goods distribution schedules

3. Problem Statement

Based on identified issues, customers experience several problems caused by unstructured systems and non-integrated business processes. These problems affect to business continuity

3.1. Customers Problem

The main problems experienced by customers include:

- Delays in order processing and delivery

  Customers often face longer waiting times due to efficient coordination between department absence of real-time and receipt scheduling.

- Inaccurate and inconsistent information

  Customers receive conflicting information regarding product availability, order status, and delivery timelines because data is stored in separate systems.

- Errors in order fulfillment

  Issues such as incorrect items, incorrect quantities, or incomplete deliveries occur due to the lack centralized control over inbound and outbound goods

- Billing and payment discrepancies
  Mismatches between orders, deliveries, and invoice

3.2. Problem Characteristics

- Open-Ended Problem

  The problems cannot be resolved by a single short-term fix, as they are interconnected across multiple business functions and require an integrated system solution

- Urgent Problem

  These issues directly impact customer satisfaction, increase complaints, and pose a risk of customer churn if not addressed promptly

- Real-World Problem

  The problems reflect common challenges faced by companies with manual or fragmented systems, where customers demand fast, accurate, and transparent services

4. Objective

The objective of this project is:

- To integrate business processes across departments
  To unify business functions such as warehouse, purchasing, and logistics into a single centralized system
- To streamline business process flow
  To create a more structured, efficient, and standardized workflow for order processing, inventory management, and delivery operations
- To support better decision-making
  To provide management with real-time, accurate, and reliable reports
  operational and strategic decision-making
- To increase operational efficiency and productivity
  To reduce manual work, human error, and process redundancy through
  System automation
- To improve inventory visibility and control
  To enable real-time tracking of inbound and outbound goods, stock levels
  and delivery schedules

B. PROBLEM ANALYSIS

1. Sustainability Aspect

The current business processes rely heavily on manual activities and disconnected systems, which are not sustainable in the long term. The lack of integration causes repeated data entry, inefficient use of resources, and difficulty in maintaining consistent operational standards as the company grows

- High dependency manual processes increases the risk of human error
- Inefficient use of time, manpower, and operational resources
- Limited scalability to support future business growth
- Difficulty in maintaining consistent and standardized processes

2. Operational Aspect

Absence of an integrated ERP system reduce overall operational efficiency and responsiveness

- Slow order-to-delivery cycle
- Delays in accessing accurate information
- Limited real-time monitoring of business activities
- Ineffective inter-department communication

3. Manufacturability Aspect

Although the company may not be a manufacturing-based organization, this aspect refers to the operability and ease of executing daily operations. The lack of standardized system makes operational activities difficult to manage and control

- Complex and inconsistent workflows across departments
- Difficult in coordinating inbound and outbound goods activities
- Limited traceability of orders, deliveries, and receipts
- High operational dependency on individual experience rather than system-driven processes

C. SOLUTION SELECTION & SCENARIOS

1. Business Process Improvement Without System Integration

This option focuses on improving the current way of working by updating standard operating procedures (SOPs), improving coordination between departments, and increasing manual supervision in daily operations. The main objective is to reduce inefficiencies by encouraging better discipline and more consistent processes, without introducing a new system.

This approach requires relatively low cost and only small changes to the existing setup. However, it does not solve the main problem, which is scattered and unconnected data. Since this solution relies heavily on human involvement, it can be difficult to maintain over time, especially as business activities and transaction volumes continue to grow.

2. Implementation of Department-Based Systems

The solution suggests using separate systems for each department, such as individual applications for sales, inventory, and finance. With dedicated systems in place, each department can work more efficiently and reduce some manual tasks.

However, because these systems are not fully connected, data is still stored separately across departments. This makes coordination more difficult and limits access to real-time information. In the long run, maintaining multiple systems and handling system integration can increase costs and reduce the effectiveness of overall business process management.

3. Implementation of an Integrated ERP System

This solution proposes the use of an integrated Enterprise Resource Planning (ERP) system that connects all core business functions within a single platform. With this system, data can be shared in real time, workflows become more structured, and coordination between departments improves significantly.

Although this option requires higher initial investment, longer implementation time, and proper user training, it provides the most long-term benefits. The ERP system supports business growth, improves decision-making, and helps deliver better service quality to customers.

4. Solution Selection

After reviewing all the available options, the integrated ERP system is considered the most suitable solution. This approach directly addresses the main issues by reducing data separation, improving process integration, and supporting long-term business sustainability. Therefore, implementing an ERP system is seen as the best strategic choice to improve overall business performance.

D. DEVELOPMENT EFFORT

1. Manpower Job Description

Job Description		
Name	**Position**	**Job Description**
Muhammad Hafid Al Asyrofi	Leader	1. Building ERD for Business Flow ERP 2. Execution Front-End 3. QA-Tester 4. UI-UX
Muhammad Kasyfil Aziz Setiadi	Member	1. Building ERD for Business Flow ERP 2. Make Back-End for logic flow 3. QA- Tester 4. UI-UX
Muhammad Habibi Dzikra	Member	1. Building ERD for Business Flow ERP 2. Make Back-End for logic flow 3. QA-Tester APP

2. Development Tools

The following tools are required for building an Integrated business system. And the tools will be support system design, implementation, testing, and deploy

- Programming Language

  For the programming language we use php v8.1 and little bit java for some logic and for the database we will use sql

- Backend development framework

  - laravel v10.1 for the main backend framework to develop a structure, secure, and scalable web base application following mvc architecture

  - composer v2 for dependency management in laravel project

- Frontend development tools
    - HTML for user interface
    - CSS use for styling and layout design
    - Bootstrap for make interactive and responsive UI

- Database management system

  For the database we will use mySQL v8.2 because we more familiar with mySQL

- Development environment & tools
    - Vscode for primary code editor
    - Xampp used as local development environment
    - Github / gitlab used for source code repository management

3. Test Equipment

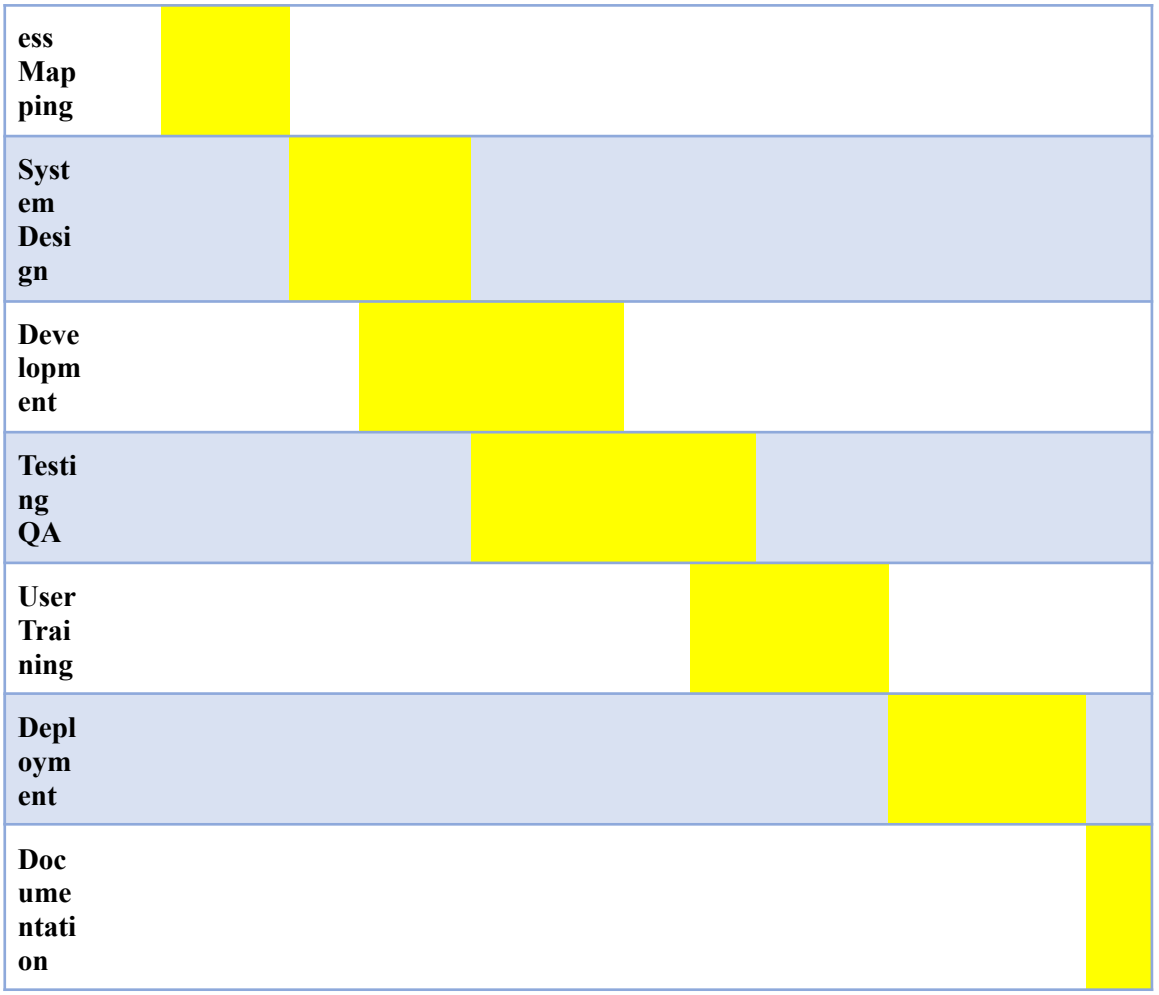
No	Equipment	Type	Specification/Function
1	Laptop	Hardware	Development and Testing
2	Smartphone	Harware	Mobile Testing
3	Web Browser	Software	Functional Testing
4	ERP Test Database	Software	System Testing With Dummy Data

4. Cost Estimation

No	Item	Description	Cost
1	VPS server	Cloud Server For ERP Deployment	Rp 200.000 - Rp 300.000 / Month
2	Domain	Application Domain	Rp 150.000 - 250.000 / Month
3	Laptop	Hardware	-

5. Timelines

Phase	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16
Requirement Analysis																
Business Proc																


ess Map ping
Syst em Desi gn
Deve lopm ent
Testi ng QA
User Trai ning
Depl oym ent
Doc ume ntati on